batmonitor 0.76-py: decision tables (bm_tables.h, 15 tables, 67 rows) - first matching row wins, - = any; green 1, red 0; share = input combinations the row decides (DEAD = never reached)

D7 house on AC-out1

D7

mode_off	mode_on	over_on	under_off	under_long	on	**on**	**under**	share
1	-	-	-	-	-	**0**	**0**	32/64
-	1	-	-	-	-	**1**	**0**	16/64
-	-	1	-	-	-	**1**	**0**	8/64
-	-	-	1	1	-	**0**	**1**	2/64
-	-	-	1	-	1	**1**	**1**	1/64
-	-	-	1	-	-	**0**	**1**	1/64
-	-	-	-	-	-	**KEEP**	**0**	4/64

D7: house on AC-out1. HOUSE_ACOUT 0 = off, 1 = on, auto: on above 99 W, off after 60 s below 30 W.

D5 DO4 pulse

D5

lockout_over	acout_over	pcc_over	grace_over	pcc_hard	**do4**	share
0	-	-	-	-	**0**	16/32
-	1	-	-	-	**1**	8/32
-	-	1	1	-	**1**	2/32
-	-	1	-	1	**1**	1/32
-	-	-	-	-	**0**	5/32

D5: DO4 pulse (FoxESS shedding): PCC > 20 kW for the grace time or > DO4_HARD_W, or grid offline and PV overfeeds one AC-out; never twice within the lockout.

D3 discharge PROP / IDLE

D3

any_dis	house_import	prop_prev	prop_hold	deficit_gt_hold	**prop**	share
0	-	-	-	-	**0**	16/32
-	1	-	-	-	**1**	8/32
-	-	1	-	1	**1**	2/32
-	-	-	1	1	**1**	1/32
-	-	-	-	-	**0**	5/32

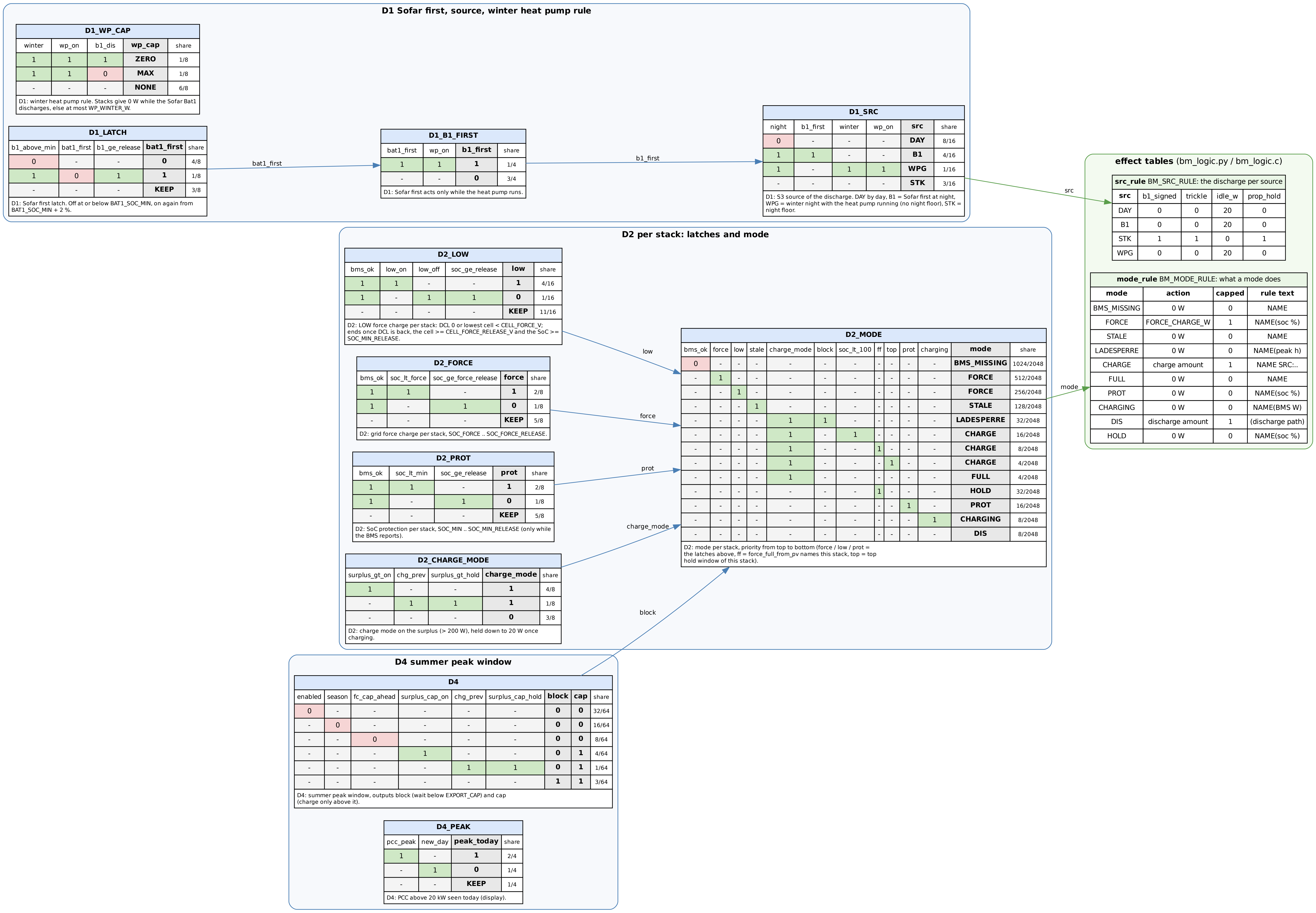
D3: PROPORTIONAL (1) while importing, or (already proportional / night floor) while the deficit > 20 W.

SoC balancing

DLEAD

both_ok	has_lead	diff_gt_on	diff_lt_off	**lead**	share
0	-	-	-	**NONE**	8/16
-	0	1	-	**NEW**	2/16
-	1	-	1	**END**	2/16
-	-	-	-	**KEEP**	4/16

SoC balancing: the stack more than SOC_BALANCE_ON ahead leads, cleared below SOC_BALANCE_OFF.



D1 Sofar first, source, winter heat pump rule

D1_WP_CAP

winter	wp_on	b1_dis	**wp_cap**	share
1	1	1	**ZERO**	1/8
1	1	0	**MAX**	1/8
-	-	-	**NONE**	6/8

D1: winter heat pump rule. Stacks give 0 W while the Sofar Bat1 discharges, else at most WP_WINTER_W.

D1_LATCH

b1_above_min	bat1_first	b1_ge_release	**bat1_first**	share
0	-	-	**0**	4/8
1	0	1	**1**	1/8
-	-	-	**KEEP**	3/8

D1: Sofar first latch. Off at or below BAT1_SOC_MIN, on again from BAT1_SOC_MIN + 2 %.

D1_B1_FIRST

bat1_first	wp_on	**b1_first**	share
1	1	**1**	1/4
-	-	**0**	3/4

D1: Sofar first acts only while the heat pump runs.

D1_SRC

night	b1_first	winter	wp_on	**src**	share
0	-	-	-	**DAY**	8/16
1	1	-	-	**B1**	4/16
1	-	1	1	**WPG**	1/16
-	-	-	-	**STK**	3/16

D1: S3 source of the discharge. DAY by day, B1 = Sofar first at night, WPG = winter night with the heat pump running (no night floor), STK = night floor.

D2 per stack: latches and mode

D2_LOW

bms_ok	low_on	low_off	soc_ge_release	**low**	share
1	1	-	-	**1**	4/16
1	-	1	1	**0**	1/16
-	-	-	-	**KEEP**	11/16

D2: LOW force charge per stack: DCL 0 or lowest cell < CELL_FORCE_V; ends once DCL is back, the cell >= CELL_FORCE_RELEASE_V and the SoC >= SOC_MIN_RELEASE.

D2_FORCE

bms_ok	soc_lt_force	soc_ge_force_release	**force**	share
1	1	-	**1**	2/8
1	-	1	**0**	1/8
-	-	-	**KEEP**	5/8

D2: grid force charge per stack, SOC_FORCE .. SOC_FORCE_RELEASE.

D2_PROT

bms_ok	soc_lt_min	soc_ge_release	**prot**	share
1	1	-	**1**	2/8
1	-	1	**0**	1/8
-	-	-	**KEEP**	5/8

D2: SoC protection per stack, SOC_MIN .. SOC_MIN_RELEASE (only while the BMS reports).

D2_CHARGE_MODE

surplus_gt_on	chg_prev	surplus_gt_hold	**charge_mode**	share
1	-	-	**1**	4/8
-	1	1	**1**	1/8
-	-	-	**0**	3/8

D2: charge mode on the surplus (> 200 W), held down to 20 W once charging.

D2_MODE

bms_ok	force	low	stale	charge_mode	block	soc_lt_100	ff	top	prot	charging	**mode**	share
0	-	-	-	-	-	-	-	-	-	-	**BMS_MISSING**	1024/2048
-	1	-	-	-	-	-	-	-	-	-	**FORCE**	512/2048
-	-	1	-	-	-	-	-	-	-	-	**FORCE**	256/2048
-	-	-	1	-	-	-	-	-	-	-	**STALE**	128/2048
-	-	-	-	1	1	-	-	-	-	-	**LADESPERRE**	32/2048
-	-	-	-	1	-	1	-	-	-	-	**CHARGE**	16/2048
-	-	-	-	1	-	-	1	-	-	-	**CHARGE**	8/2048
-	-	-	-	1	-	-	-	1	-	-	**CHARGE**	4/2048
-	-	-	-	1	-	-	-	-	-	-	**FULL**	4/2048
-	-	-	-	-	-	-	1	-	-	-	**HOLD**	32/2048
-	-	-	-	-	-	-	-	-	1	-	**PROT**	16/2048
-	-	-	-	-	-	-	-	-	-	1	**CHARGING**	8/2048
-	-	-	-	-	-	-	-	-	-	-	**DIS**	8/2048

D2: mode per stack, priority from top to bottom (force / low / prot = the latches above, ff = force_full_from_pv names this stack, top = top hold window of this stack).

D4 summer peak window

D4

enabled	season	fc_cap_ahead	surplus_cap_on	chg_prev	surplus_cap_hold	**block**	**cap**	share
0	-	-	-	-	-	**0**	**0**	32/64
-	0	-	-	-	-	**0**	**0**	16/64
-	-	0	-	-	-	**0**	**0**	8/64
-	-	-	1	-	-	**0**	**1**	4/64
-	-	-	-	1	1	**0**	**1**	1/64
-	-	-	-	-	-	**1**	**1**	3/64

D4: summer peak window, outputs block (wait below EXPORT_CAP) and cap (charge only above it).

D4_PEAK

pcc_peak	new_day	**peak_today**	share
1	-	**1**	2/4
-	1	**0**	1/4
-	-	**KEEP**	1/4

D4: PCC above 20 kW seen today (display).

effect tables (bm_logic.py / bm_logic.c)

src_rule BM_SRC_RULE: the discharge per source

src	b1_signed	trickle	idle_w	prop_hold
DAY	0	0	20	0
B1	0	0	20	0
STK	1	1	0	1
WPG	0	0	20	0

mode_rule BM_MODE_RULE: what a mode does

mode	**action**	**capped**	**rule text**
BMS_MISSING	0 W	0	NAME
FORCE	FORCE_CHARGE_W	1	NAME(soc %)
STALE	0 W	0	NAME
LADESPERRE	0 W	0	NAME(peak h)
CHARGE	charge amount	1	NAME SRC:..
FULL	0 W	0	NAME
PROT	0 W	0	NAME(soc %)
CHARGING	0 W	0	NAME(BMS W)
DIS	discharge amount	1	(discharge path)
HOLD	0 W	0	NAME(soc %)

batmonitor: observed states 08.10.2026 .. 05.11.2026 (wagodb.pv_victron, one row per minute = the state of the last 5 s cycle; generated 2026-11-05 13:34)

coverage 08.10. 11:34 .. 05.11. 13:34: 40441 minutes, 40375 logged with the rules (99.84 %), 66 logger off, 16024 with a D1 cell (bm_d1_id since 25.10.)

day	logged	logger off	D1 cells	rule states
Thu 08.10.	746	0	-	24
Fri 09.10.	1440	0	-	20
Sat 10.10.	1440	0	-	45
Sun 11.10.	1440	0	-	34
Mon 12.10.	1440	0	-	26
Tue 13.10.	1440	0	-	26
Wed 14.10.	1440	0	-	24
Thu 15.10.	1440	0	-	23
Fri 16.10.	1440	0	-	21
Sat 17.10.	1440	0	-	22
Sun 18.10.	1440	0	-	17
Mon 19.10.	1440	0	-	22
Tue 20.10.	1440	0	-	20
Wed 21.10.	1440	0	-	17
Thu 22.10.	1440	0	-	29
Fri 23.10.	1440	0	-	22
Sat 24.10.	1440	0	-	51
Sun 25.10.	1440	0	6	38
Mon 26.10.	1440	0	8	39
Tue 27.10.	1433	7	8	27
Wed 28.10.	1440	0	10	6
Thu 29.10.	1418	22	17	21
Fri 30.10.	1403	37	9	33
Sat 31.10.	1440	0	8	27
Sun 01.11.	1440	0	8	24
Mon 02.11.	1440	0	12	30
Tue 03.11.	1440	0	6	26
Wed 04.11.	1440	0	8	26
Thu 05.11.	815	0	5	16

gaps (minutes without a row)

27.10. 15:38 .. 15:43 6 min
27.10. 17:18 .. 17:18 1 min
29.10. 20:28 .. 20:49 22 min
30.10. 06:44 .. 06:46 3 min
30.10. 06:59 .. 07:23 25 min
30.10. 07:36 .. 07:36 1 min

D1 cells (pv_victron.bm_d1_id) through the D1 tables of bm_tables.h

cell	bits	minutes	set bits	src	wp_cap
93	1+4+8+16+64	4044	night, b1_above_min, b1_ge_release, winter, bat1_first	STK	none
95	1+2+4+8+16+64	3641	night, wp_on, b1_above_min, b1_ge_release, winter, bat1_first	B1	WP_WINTER_W
92	4+8+16+64	3190	b1_above_min, b1_ge_release, winter, bat1_first	DAY	none
94	2+4+8+16+64	2594	wp_on, b1_above_min, b1_ge_release, winter, bat1_first	DAY	WP_WINTER_W
127	1+2+4+8+16+32+64	575	night, wp_on, b1_above_min, b1_ge_release, winter, b1_dis, bat1_first	B1	0 W
87	1+2+4+16+64	389	night, wp_on, b1_above_min, winter, bat1_first	B1	WP_WINTER_W
85	1+4+16+64	374	night, b1_above_min, winter, bat1_first	STK	none
125	1+4+8+16+32+64	320	night, b1_above_min, b1_ge_release, winter, b1_dis, bat1_first	STK	none
126	2+4+8+16+32+64	263	wp_on, b1_above_min, b1_ge_release, winter, b1_dis, bat1_first	DAY	0 W
124	4+8+16+32+64	186	b1_above_min, b1_ge_release, winter, b1_dis, bat1_first	DAY	none
17	1+16	111	night, winter	STK	none
16	16	107	winter	DAY	none
19	1+2+16	92	night, wp_on, winter	WPG	WP_WINTER_W
18	2+16	69	wp_on, winter	DAY	WP_WINTER_W
84	4+16+64	51	b1_above_min, winter, bat1_first	DAY	none
22	2+4+16	11	wp_on, b1_above_min, winter	DAY	WP_WINTER_W
86	2+4+16+64	4	wp_on, b1_above_min, winter, bat1_first	DAY	WP_WINTER_W
28	4+8+16	1	b1_above_min, b1_ge_release, winter	DAY	none
61	1+4+8+16+32	1	night, b1_above_min, b1_ge_release, winter, b1_dis	STK	none
29	1+4+8+16	1	night, b1_above_min, b1_ge_release, winter	STK	none

rule states L1 | L2 | L3 (measured values in brackets removed), 275 different incl. logger off (no row / no rules)

minutes	%	L1 | L2 | L3
7395	18.3	PROP WP:OFF SRC:STK EQ | PROP WP:OFF SRC:STK EQ | PROP WP:OFF SRC:STK EQ
2547	6.3	PROP SRC:STK EQ | PROP SRC:STK EQ | PROP SRC:STK EQ
2358	5.8	PROP SRC:B1 EQ WPMAX(100) | PROP SRC:B1 EQ WPMAX(100) | PROP SRC:B1 EQ WPMAX(100)
1740	4.3	HOUSE SRC:Z2 EQ | HOUSE SRC:Z2 EQ | HOUSE SRC:Z2 EQ
1625	4.0	FULL | FULL | FULL
1560	3.9	CHARGE WP:OFF SRC:DAY EQ | CHARGE WP:OFF SRC:DAY EQ | CHARGE WP:OFF SRC:DAY EQ
1493	3.7	CHARGE WP:OFF SRC:DAY BAL | CHARGE WP:OFF SRC:DAY BAL | CHARGE WP:OFF SRC:DAY BAL
1364	3.4	FORCE LOW | PROT | PROT
1138	2.8	CHARGE SRC:DAY EQ | CHARGE SRC:DAY EQ | CHARGE SRC:DAY EQ
1031	2.5	PROP WP:OFF SRC:STK BAL | PROP WP:OFF SRC:STK WAIT | PROP WP:OFF SRC:STK WAIT
993	2.5	CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL
957	2.4	PROP WP:FC SRC:B1 EQ | PROP WP:FC SRC:B1 EQ | PROP WP:FC SRC:B1 EQ
877	2.2	CHARGE WP:OFF SRC:DAY EQ | FULL | FULL
723	1.8	PROP WP:FC SRC:DAY EQ | PROP WP:FC SRC:DAY EQ | PROP WP:FC SRC:DAY EQ
678	1.7	PROPORTIONAL|NIGHT|FC | PROPORTIONAL|NIGHT|FC | PROPORTIONAL|NIGHT|FC
537	1.3	CHARGE SRC:DAY EQ TOP CELL | FULL | FULL
497	1.2	IDLE WP:OFF SRC:DAY EQ | IDLE WP:OFF SRC:DAY EQ | IDLE WP:OFF SRC:DAY EQ
484	1.2	PROP SRC:DAY EQ WPMAX(100) | PROP SRC:DAY EQ WPMAX(100) | PROP SRC:DAY EQ WPMAX(100)
469	1.2	PROP WP:FC SRC:B1 BAL | PROP WP:FC SRC:B1 SPILL | PROP WP:FC SRC:B1 SPILL
461	1.1	PROP WP:OFF SRC:DAY EQ | PROP WP:OFF SRC:DAY EQ | PROP WP:OFF SRC:DAY EQ
454	1.1	PROP SRC:DAY EQ | PROP SRC:DAY EQ | PROP SRC:DAY EQ
440	1.1	PROP WP:OFF SRC:STK WAIT | PROP WP:OFF SRC:STK BAL | PROP WP:OFF SRC:STK BAL
430	1.1	STACK1_MUST:PV_SURPLUS|CHARGE | STACK2_PYTES:PV_SURPLUS|CHARGE | STACK2_PYTES:PV_SURPLUS|CHARGE
391	1.0	FORCE LOW | FORCE LOW | FORCE LOW
360	0.9	PROPORTIONAL|NIGHT | PROPORTIONAL|NIGHT | PROPORTIONAL|NIGHT
352	0.9	PROP SRC:B1 EQ WPMAX(0) | PROP SRC:B1 EQ WPMAX(0) | PROP SRC:B1 EQ WPMAX(0)
301	0.7	HOUSE SRC:Z2 BAL | HOUSE SRC:Z2 WAIT | HOUSE SRC:Z2 WAIT
290	0.7	CHARGE WP:OFF SRC:DAY BAL CCL | FULL | FULL
275	0.7	PROP SRC:B1 BAL WPMAX(100) | PROP SRC:B1 WAIT WPMAX(100) | PROP SRC:B1 WAIT WPMAX(100)
263	0.7	PROP SRC:DAY BAL WPMAX(100) | PROP SRC:DAY WAIT WPMAX(100) | PROP SRC:DAY WAIT WPMAX(100)
246	0.6	PROP SRC:B1 WAIT WPMAX(100) | PROP SRC:B1 BAL WPMAX(100) | PROP SRC:B1 BAL WPMAX(100)
245	0.6	CHARGE WP:FC SRC:DAY EQ | CHARGE WP:FC SRC:DAY EQ | CHARGE WP:FC SRC:DAY EQ
242	0.6	BMS_MISSING | PROT | PROT
237	0.6	CHARGE SRC:Z2 BAL | CHARGE SRC:Z2 BAL | CHARGE SRC:Z2 BAL
237	0.6	CHARGE SRC:DAY EQ TOP | FULL | FULL
202	0.5	PROP SRC:B1 1PH WPMAX(1000) | PROP SRC:B1 STBY WPMAX(1000) | PROP SRC:B1 STBY WPMAX(1000)
197	0.5	PROP SRC:STK 1PH | PROP SRC:STK STBY | PROP SRC:STK STBY
190	0.5	STACK1_MUST:PV_SURPLUS | STACK2_PYTES:PV_SURPLUS | STACK2_PYTES:PV_SURPLUS
183	0.5	PROP SRC:STK BAL | PROP SRC:STK WAIT | PROP SRC:STK WAIT
178	0.4	PROP SRC:STK STBY | PROP SRC:STK 1PH | PROP SRC:STK STBY
176	0.4	IDLE WP:OFF SRC:DAY BAL | IDLE WP:OFF SRC:DAY WAIT | IDLE WP:OFF SRC:DAY WAIT
173	0.4	PROP SRC:B1 STBY WPMAX(1000) | PROP SRC:B1 1PH WPMAX(1000) | PROP SRC:B1 STBY WPMAX(1000)
170	0.4	PROP WP:FC SRC:DAY BAL | PROP WP:FC SRC:DAY WAIT | PROP WP:FC SRC:DAY WAIT
169	0.4	PROP SRC:STK WAIT | PROP SRC:STK BAL | PROP SRC:STK BAL
140	0.3	PROP SRC:B1 EQ | PROP SRC:B1 EQ | PROP SRC:B1 EQ
136	0.3	PROP WP:OFF SRC:DAY BAL | PROP WP:OFF SRC:DAY WAIT | PROP WP:OFF SRC:DAY WAIT
126	0.3	IDLE SRC:DAY EQ | IDLE SRC:DAY EQ | IDLE SRC:DAY EQ
119	0.3	PROP SRC:STK EQ WPLOW | PROP SRC:STK EQ WPLOW | PROP SRC:STK EQ WPLOW
114	0.3	CHARGE WP:OFF SRC:DAY BAL | FULL | FULL
108	0.3	PROP WP:FC SRC:B1 BAL | PROP WP:FC SRC:B1 WAIT | PROP WP:FC SRC:B1 WAIT
100	0.2	PROP SRC:B1 EQ WPLOW | PROP SRC:B1 EQ WPLOW | PROP SRC:B1 EQ WPLOW
98	0.2	IDLE SRC:DAY BAL WPLOW | IDLE SRC:DAY WAIT WPLOW | IDLE SRC:DAY WAIT WPLOW
95	0.2	HOUSE SRC:Z2 BAL | PROT | PROT
94	0.2	FORCE LOW | CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL
93	0.2	FORCE LOW | PROP SRC:STK EQ | PROP SRC:STK EQ
93	0.2	CHARGE SRC:Z2 EQ | CHARGE SRC:Z2 EQ | CHARGE SRC:Z2 EQ
91	0.2	IDLE SRC:DAY EQ WPMAX(0) | IDLE SRC:DAY EQ WPMAX(0) | IDLE SRC:DAY EQ WPMAX(0)
91	0.2	FORCE LOW | HOUSE SRC:Z2 EQ | HOUSE SRC:Z2 EQ
90	0.2	PROP WP:T1900 SRC:B1 EQ | PROP WP:T1900 SRC:B1 EQ | PROP WP:T1900 SRC:B1 EQ
89	0.2	HOUSE SRC:Z2 STBY | HOUSE SRC:Z2 1PH | HOUSE SRC:Z2 STBY
85	0.2	CHARGE WP:OFF SRC:DAY BAL | FORCE LOW | FORCE LOW
84	0.2	PROP SRC:DAY BAL | PROP SRC:DAY WAIT | PROP SRC:DAY WAIT
79	0.2	PROP SRC:DAY EQ WPMAX(1000) | PROP SRC:DAY EQ WPMAX(1000) | PROP SRC:DAY EQ WPMAX(1000)
77	0.2	STACK1_MUST:PV_SURPLUS|CHARGE | STACK2_PYTES:PV_SURPLUS | STACK2_PYTES:PV_SURPLUS
76	0.2	PROPORTIONAL|NIGHT|WP1900 | PROPORTIONAL|NIGHT|WP1900 | PROPORTIONAL|NIGHT|WP1900
75	0.2	PROP SRC:DAY BAL WPMAX(1000) | PROP SRC:DAY WAIT WPMAX(1000) | PROP SRC:DAY WAIT WPMAX(1000)
71	0.2	CHARGE SRC:DAY BAL TOP CELL | FULL | FULL
66	0.2	PROP SRC:B1 BAL WPMAX(1000) | PROP SRC:B1 WAIT WPMAX(1000) | PROP SRC:B1 WAIT WPMAX(1000)
66	0.2	HOUSE SRC:Z2 WAIT | HOUSE SRC:Z2 BAL | HOUSE SRC:Z2 BAL
66	0.2	logger off
65	0.2	BMS_MISSING | FULL | FULL
64	0.2	STACK1_MUST:BMS_MISSING | IDLE|NIGHT|FC | IDLE|NIGHT|FC
64	0.2	CHARGE WP:FC SRC:DAY BAL | CHARGE WP:FC SRC:DAY BAL | CHARGE WP:FC SRC:DAY BAL
62	0.2	PROPORTIONAL|FC | PROPORTIONAL|FC | PROPORTIONAL|FC
61	0.2	PROP SRC:DAY EQ WPMAX(0) | PROP SRC:DAY EQ WPMAX(0) | PROP SRC:DAY EQ WPMAX(0)
55	0.1	IDLE WP:FCBAD SRC:B1 EQ | IDLE WP:FCBAD SRC:B1 EQ | IDLE WP:FCBAD SRC:B1 EQ
55	0.1	PROP SRC:DAY 1PH | PROP SRC:DAY STBY | PROP SRC:DAY STBY
54	0.1	PROP SRC:B1 WAIT WPMAX(0) | PROP SRC:B1 BAL WPMAX(0) | PROP SRC:B1 BAL WPMAX(0)
52	0.1	IDLE WP:OFF SRC:STK EQ | IDLE WP:OFF SRC:STK EQ | IDLE WP:OFF SRC:STK EQ
52	0.1	FULL | CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL
47	0.1	CHARGE SRC:DAY BAL CCL | FULL | FULL
46	0.1	PROP WP:T1900 SRC:DAY EQ | PROP WP:T1900 SRC:DAY EQ | PROP WP:T1900 SRC:DAY EQ
46	0.1	PROP SRC:B1 EQ WPMAX(1000) | PROP SRC:B1 EQ WPMAX(1000) | PROP SRC:B1 EQ WPMAX(1000)
46	0.1	CHARGE SRC:DAY BAL | FULL | FULL
45	0.1	PROP WP:FC SRC:DAY WAIT | PROP WP:FC SRC:DAY BAL | PROP WP:FC SRC:DAY BAL
44	0.1	IDLE|NIGHT|FC | IDLE|NIGHT|FC | IDLE|NIGHT|FC
42	0.1	IDLE WP:FC SRC:DAY EQ | IDLE WP:FC SRC:DAY EQ | IDLE WP:FC SRC:DAY EQ
42	0.1	CHARGE SRC:Z2 BAL CCL | FULL | FULL
41	0.1	STACK1_MUST:BMS_MISSING | STACK2_PYTES:PV_SURPLUS|CHARGE | STACK2_PYTES:PV_SURPLUS|CHARGE
41	0.1	CHARGE SRC:Z2 BAL | FULL | FULL
37	0.1	PROPORTIONAL|NIGHT|BALANCE_HOLD | PROPORTIONAL|NIGHT|BALANCE | PROPORTIONAL|NIGHT|BALANCE
37	0.1	PROP WP:FC SRC:B1 WAIT | PROP WP:FC SRC:B1 BAL | PROP WP:FC SRC:B1 BAL
36	0.1	FORCE LOW | PROP SRC:B1 EQ WPMAX(1000) | PROP SRC:B1 EQ WPMAX(1000)
35	0.1	CHARGE WP:Z2 SRC:DAY EQ | CHARGE WP:Z2 SRC:DAY EQ | CHARGE WP:Z2 SRC:DAY EQ
34	0.1	PROP SRC:DAY STBY | PROP SRC:DAY 1PH | PROP SRC:DAY STBY
33	0.1	STACK1_MUST:BMS_MISSING | IDLE|FC | IDLE|FC
33	0.1	PROP WP:OFF SRC:DAY BAL | FORCE LOW | FORCE LOW
33	0.1	PROP WP:OFF SRC:STK BAL | FORCE LOW | FORCE LOW
32	0.1	PROP WP:OFF SRC:STK BAL | PROT | PROT
31	0.1	IDLE|FC|BALANCE | IDLE|FC|BALANCE_HOLD | IDLE|FC|BALANCE_HOLD
31	0.1	PROP SRC:DAY STBY WPMAX(1000) | PROP SRC:DAY 1PH WPMAX(1000) | PROP SRC:DAY STBY WPMAX(1000)
29	0.1	PROP WP:LOWSOC SRC:B1 EQ | PROP WP:LOWSOC SRC:B1 EQ | PROP WP:LOWSOC SRC:B1 EQ
27	0.1	IDLE WP:FCBAD SRC:DAY EQ | IDLE WP:FCBAD SRC:DAY EQ | IDLE WP:FCBAD SRC:DAY EQ
26	0.1	PROPORTIONAL|NIGHT|FC|BALANCE_HOLD | PROPORTIONAL|NIGHT|FC|BALANCE | PROPORTIONAL|NIGHT|FC|BALANCE
26	0.1	EXPCAP SRC:Z2 EQ | EXPCAP SRC:Z2 EQ | EXPCAP SRC:Z2 EQ
24	0.1	PROP SRC:DAY 1PH WPMAX(1000) | PROP SRC:DAY STBY WPMAX(1000) | PROP SRC:DAY STBY WPMAX(1000)
23	0.1	BMS_MISSING | CHARGE SRC:DAY EQ | CHARGE SRC:DAY EQ
23	0.1	CHARGE SRC:Z2 BAL | FORCE LOW | FORCE LOW
22	0.1	IDLE|BALANCE | IDLE|BALANCE_HOLD | IDLE|BALANCE_HOLD
22	0.1	IDLE|Z2 | IDLE|Z2 | IDLE|Z2
22	0.1	IDLE|FC | IDLE|FC | IDLE|FC
22	0.1	PROP WP:T1900 SRC:B1 BAL | PROP WP:T1900 SRC:B1 WAIT | PROP WP:T1900 SRC:B1 WAIT
22	0.1	IDLE WP:OFF SRC:DAY BAL | FORCE LOW | FORCE LOW
21	0.1	CHARGING | CHARGING | CHARGING
21	0.1	PROP SRC:B1 BAL WPLOW | PROP SRC:B1 WAIT WPLOW | PROP SRC:B1 WAIT WPLOW
21	0.1	PROP SRC:STK BAL | FORCE LOW | FORCE LOW
21	0.1	CHARGE SRC:DAY EQ CELL | FULL | FULL
20	0.0	STACK2_PYTES:PV_SURPLUS|CHARGE | STACK1_MUST:PV_SURPLUS|CHARGE | STACK2_PYTES:PV_SURPLUS|CHARGE
20	0.0	PROP SRC:DAY BAL WPLOW | PROP SRC:DAY WAIT WPLOW | PROP SRC:DAY WAIT WPLOW
20	0.0	IDLE SRC:DAY BAL | IDLE SRC:DAY WAIT | IDLE SRC:DAY WAIT
20	0.0	CHARGE SRC:DAY FF CCL | FULL | FULL
20	0.0	HOUSE SRC:Z2 BAL | FORCE LOW | FORCE LOW
19	0.0	PROPORTIONAL | PROPORTIONAL | PROPORTIONAL
19	0.0	CHARGING | PROP SRC:DAY EQ WPMAX(100) | PROP SRC:DAY EQ WPMAX(100)
19	0.0	IDLE SRC:DAY EQ WPMAX(100) | IDLE SRC:DAY EQ WPMAX(100) | IDLE SRC:DAY EQ WPMAX(100)
18	0.0	IDLE WP:T1900 SRC:B1 EQ | IDLE WP:T1900 SRC:B1 EQ | IDLE WP:T1900 SRC:B1 EQ
18	0.0	IDLE WP:T1900 SRC:DAY EQ | IDLE WP:T1900 SRC:DAY EQ | IDLE WP:T1900 SRC:DAY EQ
18	0.0	FORCE LOW | PROP SRC:B1 EQ WPLOW | PROP SRC:B1 EQ WPLOW
18	0.0	FORCE LOW | PROP SRC:DAY EQ WPLOW | PROP SRC:DAY EQ WPLOW
18	0.0	CHARGE SRC:Z2 BAL | CHARGE SRC:Z2 BAL CCL | CHARGE SRC:Z2 BAL CCL
18	0.0	CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ | CHARGE SRC:DAY EQ
17	0.0	IDLE WP:FC SRC:DAY BAL | IDLE WP:FC SRC:DAY WAIT | IDLE WP:FC SRC:DAY WAIT
17	0.0	IDLE SRC:DAY BAL WPMAX(0) | IDLE SRC:DAY WAIT WPMAX(0) | IDLE SRC:DAY WAIT WPMAX(0)
16	0.0	PROP WP:FC SRC:B1 BAL | PROT | PROT
15	0.0	PROP WP:FC SRC:DAY BAL | PROP WP:FC SRC:DAY SPILL | PROP WP:FC SRC:DAY SPILL
14	0.0	PROP WP:FCBAD SRC:B1 EQ | PROP WP:FCBAD SRC:B1 EQ | PROP WP:FCBAD SRC:B1 EQ
14	0.0	IDLE WP:OFF SRC:DAY BAL | CHARGING | CHARGING
14	0.0	CHARGE WP:OFF SRC:DAY EQ CCL | CHARGE WP:OFF SRC:DAY EQ | CHARGE WP:OFF SRC:DAY EQ
14	0.0	FORCE LOW | PROP WP:LOWSOC SRC:B1 EQ | PROP WP:LOWSOC SRC:B1 EQ
14	0.0	IDLE SRC:DAY BAL WPMAX(100) | IDLE SRC:DAY WAIT WPMAX(100) | IDLE SRC:DAY WAIT WPMAX(100)
13	0.0	PROPORTIONAL|FC|BALANCE | PROPORTIONAL|FC|BALANCE_HOLD | PROPORTIONAL|FC|BALANCE_HOLD
13	0.0	IDLE SRC:DAY BAL | CHARGING | CHARGING
13	0.0	PROP SRC:B1 1PH WPMAX(0) | PROP SRC:B1 STBY WPMAX(0) | PROP SRC:B1 STBY WPMAX(0)
12	0.0	CHARGING | PROP SRC:B1 EQ WPMAX(100) | PROP SRC:B1 EQ WPMAX(100)
12	0.0	CHARGE SRC:DAY EQ | FULL | FULL
11	0.0	PROP WP:OFF SRC:DAY BAL | CHARGING | CHARGING
11	0.0	BMS_MISSING | FORCE LOW | FORCE LOW
11	0.0	CHARGE WP:LOWSOC SRC:DAY BAL | CHARGE WP:LOWSOC SRC:DAY BAL | CHARGE WP:LOWSOC SRC:DAY BAL
11	0.0	FORCE LOW | CHARGE SRC:DAY EQ | CHARGE SRC:DAY EQ
11	0.0	BMS_MISSING | BMS_MISSING | BMS_MISSING
10	0.0	CHARGING | IDLE WP:OFF SRC:DAY EQ | IDLE WP:OFF SRC:DAY EQ
10	0.0	CHARGING | IDLE SRC:DAY EQ | IDLE SRC:DAY EQ
10	0.0	PROP SRC:B1 BAL | PROP SRC:B1 SPILL | PROP SRC:B1 SPILL
10	0.0	IDLE SRC:B1 EQ WPLOW | IDLE SRC:B1 EQ WPLOW | IDLE SRC:B1 EQ WPLOW
10	0.0	PROP SRC:STK BAL WPLOW | PROP SRC:STK WAIT WPLOW | PROP SRC:STK WAIT WPLOW
10	0.0	PROP SRC:DAY BAL | CHARGING | CHARGING
10	0.0	PROP SRC:B1 BAL WPMAX(1000) | FORCE LOW | FORCE LOW
9	0.0	IDLE | IDLE | IDLE
9	0.0	BMS_MISSING | CHARGING | CHARGING
9	0.0	IDLE SRC:DAY WAIT WPMAX(100) | IDLE SRC:DAY BAL WPMAX(100) | IDLE SRC:DAY BAL WPMAX(100)
8	0.0	STACK1_MUST:PV_SURPLUS|CHARGE|CCL | STACK2_PYTES:PV_SURPLUS | STACK2_PYTES:PV_SURPLUS
8	0.0	CHARGE WP:T1900 SRC:DAY EQ | CHARGE WP:T1900 SRC:DAY EQ | CHARGE WP:T1900 SRC:DAY EQ
7	0.0	CHARGE SRC:STK EQ | CHARGE SRC:STK EQ | CHARGE SRC:STK EQ
7	0.0	CHARGE SRC:DAY EQ TOP CCL | FULL | FULL
6	0.0	IDLE|WP1900 | IDLE|WP1900 | IDLE|WP1900
6	0.0	STACK1_MUST:CHARGING | IDLE|FC | IDLE|FC
6	0.0	CHARGE WP:OFF SRC:STK EQ | CHARGE WP:OFF SRC:STK EQ | CHARGE WP:OFF SRC:STK EQ
6	0.0	PROP SRC:DAY WAIT | PROP SRC:DAY BAL | PROP SRC:DAY BAL
6	0.0	CHARGE SRC:DAY FF | FULL | FULL
6	0.0	CHARGE SRC:DAY EQ TOP CCL | CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ CCL
5	0.0	STACK1_MUST:BMS_MISSING | PROPORTIONAL|NIGHT|FC | PROPORTIONAL|NIGHT|FC
5	0.0	PROP WP:T1900 SRC:DAY BAL | PROP WP:T1900 SRC:DAY WAIT | PROP WP:T1900 SRC:DAY WAIT
5	0.0	CHARGE WP:OFF SRC:STK BAL | CHARGE WP:OFF SRC:STK BAL | CHARGE WP:OFF SRC:STK BAL
5	0.0	IDLE WP:OFF SRC:STK BAL | IDLE WP:OFF SRC:STK WAIT | IDLE WP:OFF SRC:STK WAIT
5	0.0	PROP WP:OFF SRC:STK BAL | PROP WP:OFF SRC:STK SPILL | PROP WP:OFF SRC:STK SPILL
5	0.0	PROP WP:Z2 SRC:B1 EQ | PROP WP:Z2 SRC:B1 EQ | PROP WP:Z2 SRC:B1 EQ
5	0.0	CHARGE WP:LOWSOC SRC:DAY EQ | CHARGE WP:LOWSOC SRC:DAY EQ | CHARGE WP:LOWSOC SRC:DAY EQ
5	0.0	FORCE LOW | PROP SRC:DAY EQ | PROP SRC:DAY EQ
5	0.0	CHARGE SRC:DAY BAL TOP CCL | FULL | FULL
5	0.0	IDLE SRC:B1 EQ WPMAX(0) | IDLE SRC:B1 EQ WPMAX(0) | IDLE SRC:B1 EQ WPMAX(0)
5	0.0	PROP SRC:B1 EQ WPMAX(100) | CHARGING | CHARGING
4	0.0	STACK1_MUST:BMS_MISSING | PROPORTIONAL|FC | PROPORTIONAL|FC
4	0.0	PROPORTIONAL|NIGHT|WP1900|FCBAD | PROPORTIONAL|NIGHT|WP1900|FCBAD | PROPORTIONAL|NIGHT|WP1900|FCBAD
4	0.0	PROP WP:OFF SRC:DAY BAL | PROP WP:OFF SRC:DAY SPILL | PROP WP:OFF SRC:DAY SPILL
4	0.0	CHARGING | PROP WP:OFF SRC:DAY EQ | PROP WP:OFF SRC:DAY EQ
4	0.0	FORCE LOW | PROP WP:FC SRC:B1 EQ | PROP WP:FC SRC:B1 EQ
4	0.0	CHARGE WP:OFF SRC:DAY BAL | CHARGE WP:OFF SRC:DAY BAL CCL | CHARGE WP:OFF SRC:DAY BAL CCL
4	0.0	IDLE WP:OFF SRC:STK WAIT | IDLE WP:OFF SRC:STK BAL | IDLE WP:OFF SRC:STK BAL
4	0.0	CHARGE WP:OFF SRC:DAY BAL CCL | CHARGE WP:OFF SRC:DAY BAL | CHARGE WP:OFF SRC:DAY BAL
4	0.0	PROP WP:FC SRC:DAY EQ | CHARGING | CHARGING
4	0.0	PROP SRC:DAY BAL WPMAX(1000) | FORCE LOW | FORCE LOW
4	0.0	IDLE SRC:STK EQ | IDLE SRC:STK EQ | IDLE SRC:STK EQ
4	0.0	IDLE SRC:DAY 1PH | IDLE SRC:DAY STBY | IDLE SRC:DAY STBY
4	0.0	PROP SRC:DAY WAIT WPMAX(100) | PROP SRC:DAY BAL WPMAX(100) | PROP SRC:DAY BAL WPMAX(100)
4	0.0	CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ CCL
3	0.0	PROPORTIONAL|NIGHT|WP1900|BALANCE | PROPORTIONAL|NIGHT|WP1900|BALANCE_HOLD | PROPORTIONAL|NIGHT|WP1900|BALANCE_HOLD
3	0.0	CHARGE WP:T1900 SRC:B1 EQ | CHARGE WP:T1900 SRC:B1 EQ | CHARGE WP:T1900 SRC:B1 EQ
3	0.0	CHARGING | PROP WP:OFF SRC:STK EQ | PROP WP:OFF SRC:STK EQ
3	0.0	IDLE WP:FCBAD SRC:DAY BAL | IDLE WP:FCBAD SRC:DAY WAIT | IDLE WP:FCBAD SRC:DAY WAIT
3	0.0	CHARGE WP:FCBAD SRC:DAY EQ | CHARGE WP:FCBAD SRC:DAY EQ | CHARGE WP:FCBAD SRC:DAY EQ
3	0.0	PROP WP:FCBAD SRC:DAY EQ | PROP WP:FCBAD SRC:DAY EQ | PROP WP:FCBAD SRC:DAY EQ
3	0.0	IDLE WP:FC SRC:B1 EQ | IDLE WP:FC SRC:B1 EQ | IDLE WP:FC SRC:B1 EQ
3	0.0	CHARGING | FORCE LOW | FORCE LOW
3	0.0	FORCE LOW | CHARGING | CHARGING
3	0.0	PROP SRC:DAY BAL WPMAX(0) | PROP SRC:DAY WAIT WPMAX(0) | PROP SRC:DAY WAIT WPMAX(0)
3	0.0	IDLE SRC:DAY BAL WPMAX(1000) | IDLE SRC:DAY WAIT WPMAX(1000) | IDLE SRC:DAY WAIT WPMAX(1000)
3	0.0	IDLE SRC:DAY WAIT WPMAX(1000) | IDLE SRC:DAY BAL WPMAX(1000) | IDLE SRC:DAY BAL WPMAX(1000)
3	0.0	IDLE SRC:DAY STBY | IDLE SRC:DAY 1PH | IDLE SRC:DAY STBY
3	0.0	PROP SRC:DAY 1PH WPMAX(0) | PROP SRC:DAY STBY WPMAX(0) | PROP SRC:DAY STBY WPMAX(0)
2	0.0	PROPORTIONAL|BALANCE | PROPORTIONAL|BALANCE_HOLD | PROPORTIONAL|BALANCE_HOLD
2	0.0	PROPORTIONAL|WP_CAP|BALANCE | PROPORTIONAL|WP_CAP|BALANCE_HOLD | PROPORTIONAL|WP_CAP|BALANCE_HOLD
2	0.0	IDLE|FC|BALANCE | STACK2_PYTES:CHARGING | STACK2_PYTES:CHARGING
2	0.0	STACK1_MUST:PV_SURPLUS | STACK2_PYTES:PV_SURPLUS|CHARGE | STACK2_PYTES:PV_SURPLUS|CHARGE
2	0.0	PROP WP:ALL SRC:B1 EQ | PROP WP:ALL SRC:B1 EQ | PROP WP:ALL SRC:B1 EQ
2	0.0	PROP WP:FC SRC:DAY SPILL | PROP WP:FC SRC:DAY BAL | PROP WP:FC SRC:DAY BAL
2	0.0	IDLE WP:FC SRC:DAY BAL | CHARGING | CHARGING
2	0.0	CHARGE WP:FC SRC:B1 EQ | CHARGE WP:FC SRC:B1 EQ | CHARGE WP:FC SRC:B1 EQ
2	0.0	PROP WP:FC SRC:DAY BAL | CHARGING | CHARGING
2	0.0	CHARGING | PROP WP:FC SRC:DAY EQ | PROP WP:FC SRC:DAY EQ
2	0.0	CHARGE WP:LOWSOC SRC:B1 EQ | CHARGE WP:LOWSOC SRC:B1 EQ | CHARGE WP:LOWSOC SRC:B1 EQ
2	0.0	FORCE LOW | CHARGE WP:LOWSOC SRC:B1 BAL | CHARGE WP:LOWSOC SRC:B1 BAL
2	0.0	PROP SRC:DAY EQ | CHARGING | CHARGING
2	0.0	PROP SRC:B1 BAL | PROP SRC:B1 WAIT | PROP SRC:B1 WAIT
2	0.0	FORCE LOW | IDLE SRC:B1 EQ WPLOW | IDLE SRC:B1 EQ WPLOW
2	0.0	IDLE SRC:DAY BAL WPLOW | CHARGING | CHARGING
2	0.0	IDLE SRC:DAY WAIT WPMAX(0) | IDLE SRC:DAY BAL WPMAX(0) | IDLE SRC:DAY BAL WPMAX(0)
2	0.0	BMS_MISSING | HOUSE SRC:Z2 EQ | HOUSE SRC:Z2 EQ
2	0.0	PROP SRC:DAY BAL WPMAX(100) | CHARGING | CHARGING
2	0.0	CHARGING | PROP SRC:DAY EQ | PROP SRC:DAY EQ
2	0.0	PROP SRC:DAY EQ WPMAX(100) | CHARGING | CHARGING
2	0.0	CHARGE SRC:DAY BAL CCL | CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL
2	0.0	IDLE SRC:STK WAIT | IDLE SRC:STK BAL | IDLE SRC:STK BAL
1	0.0	IDLE|BALANCE | STACK2_PYTES:CHARGING | STACK2_PYTES:CHARGING
1	0.0	PROPORTIONAL|BALANCE | STACK2_PYTES:CHARGING | STACK2_PYTES:CHARGING
1	0.0	STACK1_MUST:BMS_MISSING | STACK2_PYTES:BMS_MISSING | STACK2_PYTES:BMS_MISSING
1	0.0	PROPORTIONAL|FC|BALANCE | PROPORTIONAL|FC|BALANCE_SPILL | PROPORTIONAL|FC|BALANCE_SPILL
1	0.0	IDLE|NIGHT | IDLE|NIGHT | IDLE|NIGHT
1	0.0	PROPORTIONAL|NIGHT|B1FIRST | PROPORTIONAL|NIGHT|B1FIRST | PROPORTIONAL|NIGHT|B1FIRST
1	0.0	PROP WP:FCBAD SRC:DAY BAL | PROP WP:FCBAD SRC:DAY WAIT | PROP WP:FCBAD SRC:DAY WAIT
1	0.0	CHARGE WP:FCBAD SRC:DAY BAL | CHARGE WP:FCBAD SRC:DAY BAL | CHARGE WP:FCBAD SRC:DAY BAL
1	0.0	IDLE WP:T1900 SRC:DAY BAL | IDLE WP:T1900 SRC:DAY WAIT | IDLE WP:T1900 SRC:DAY WAIT
1	0.0	PROP WP:T1900 SRC:DAY BAL | PROP WP:T1900 SRC:DAY SPILL | PROP WP:T1900 SRC:DAY SPILL
1	0.0	PROP WP:T1900 SRC:B1 BAL | PROP WP:T1900 SRC:B1 SPILL | PROP WP:T1900 SRC:B1 SPILL
1	0.0	BMS_MISSING | CHARGE WP:T1900 SRC:B1 EQ | CHARGE WP:T1900 SRC:B1 EQ
1	0.0	PROP WP:FC SRC:B1 SPILL | PROP WP:FC SRC:B1 BAL | PROP WP:FC SRC:B1 BAL
1	0.0	IDLE WP:OFF SRC:STK BAL | PROT | PROT
1	0.0	CHARGING | IDLE WP:FC SRC:DAY EQ | IDLE WP:FC SRC:DAY EQ
1	0.0	IDLE WP:FC SRC:DAY EQ | CHARGING | CHARGING
1	0.0	PROP WP:OFF SRC:DAY EQ | CHARGING | CHARGING
1	0.0	IDLE WP:NA SRC:DAY EQ | IDLE WP:NA SRC:DAY EQ | IDLE WP:NA SRC:DAY EQ
1	0.0	IDLE WP:FC SRC:B1 BAL | IDLE WP:FC SRC:B1 WAIT | IDLE WP:FC SRC:B1 WAIT
1	0.0	CHARGING | IDLE WP:LOWSOC SRC:B1 EQ | IDLE WP:LOWSOC SRC:B1 EQ
1	0.0	CHARGING | PROP SRC:DAY EQ WPLOW | PROP SRC:DAY EQ WPLOW
1	0.0	CHARGING | PROP SRC:DAY EQ WPMAX(1000) | PROP SRC:DAY EQ WPMAX(1000)
1	0.0	IDLE SRC:STK BAL | IDLE SRC:STK WAIT | IDLE SRC:STK WAIT
1	0.0	FORCE LOW | IDLE SRC:DAY EQ | IDLE SRC:DAY EQ
1	0.0	IDLE SRC:DAY BAL WPMAX(0) | CHARGING | CHARGING
1	0.0	IDLE SRC:STK 1PH | IDLE SRC:STK STBY | IDLE SRC:STK STBY
1	0.0	FULL | BMS_MISSING | BMS_MISSING
1	0.0	HOUSE SRC:Z2 BAL | CHARGING | CHARGING
1	0.0	CHARGING | HOUSE SRC:Z2 EQ | HOUSE SRC:Z2 EQ
1	0.0	CHARGE SRC:Z2 EQ | FULL | FULL
1	0.0	PROP SRC:STK EQ | BMS_MISSING | BMS_MISSING
1	0.0	CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL CCL | CHARGE SRC:DAY BAL CCL
1	0.0	PROP SRC:STK BAL | PROP SRC:STK SPILL | PROP SRC:STK SPILL
1	0.0	IDLE SRC:B1 BAL WPMAX(0) | IDLE SRC:B1 WAIT WPMAX(0) | IDLE SRC:B1 WAIT WPMAX(0)
1	0.0	PROP SRC:STK BAL | CHARGING | CHARGING
1	0.0	PROP SRC:B1 EQ WPMAX(0) | CHARGING | CHARGING
1	0.0	CHARGING | IDLE SRC:DAY EQ WPMAX(100) | IDLE SRC:DAY EQ WPMAX(100)
1	0.0	IDLE SRC:B1 WAIT WPMAX(0) | IDLE SRC:B1 BAL WPMAX(0) | IDLE SRC:B1 BAL WPMAX(0)
1	0.0	IDLE SRC:DAY EQ WPMAX(0) | CHARGING | CHARGING
1	0.0	CHARGING | PROP SRC:DAY EQ WPMAX(0) | PROP SRC:DAY EQ WPMAX(0)
1	0.0	IDLE SRC:DAY EQ | CHARGING | CHARGING
1	0.0	CHARGE SRC:DAY EQ TOP CELL | CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ CCL
1	0.0	CHARGE SRC:DAY EQ CCL | FULL | FULL